Industrial Internship Report on

QUIZ Game Using python

Prepared by

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Executive Summary
This report presents the work completed during the six-week Industrial Internship conducted by upskill Campus in collaboration with UniConverge Technologies Pvt. Ltd. (UCT). The internship focused on developing practical programming skills through a real-world project. My project, Quiz Game using Python, is a console-based application developed to test users' knowledge by asking multiple-choice questions and calculating their score. The project was implemented using Python and helped me understand programming logic, conditional statements, loops, functions, and user interaction. This internship improved my coding skills, problem-solving ability, and confidence in software development. It was a valuable learning experience that prepared me for future technical projects and career opportunities.

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Preface

Industrial internships play an important role in improving students' practical knowledge and technical skills. They help students understand how theoretical concepts are applied in real-world projects.

During this six-week internship, I worked on the project "Quiz Game using Python." The project involved designing and developing a simple quiz application where users answer multiple-choice questions and receive their final score.

This internship gave me the opportunity to improve my Python programming skills, understand software development concepts, and practice logical thinking while solving problems.

I sincerely thank upskill Campus, UniConverge Technologies Pvt. Ltd., my mentors, and Gnanamani College of Technology for providing me with this excellent learning opportunity and continuous guidance throughout the internship.

Introduction

About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

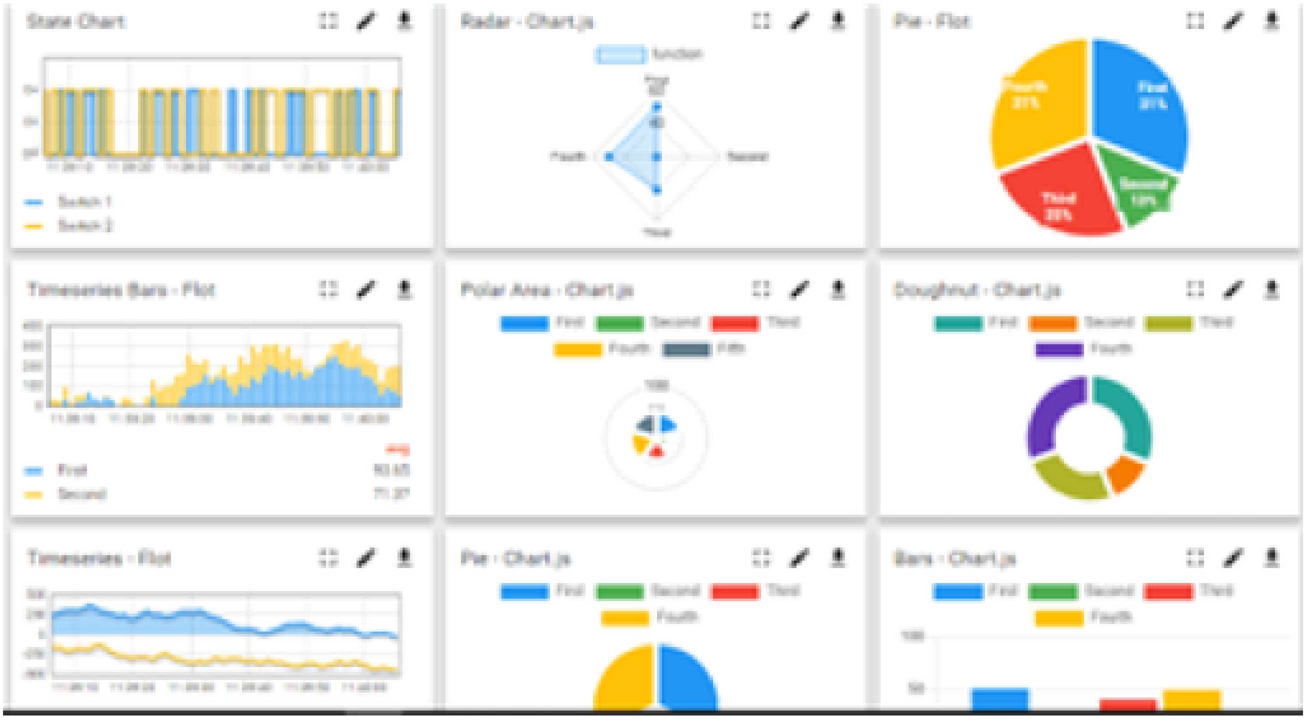
i. **UCT IoT Platform (**

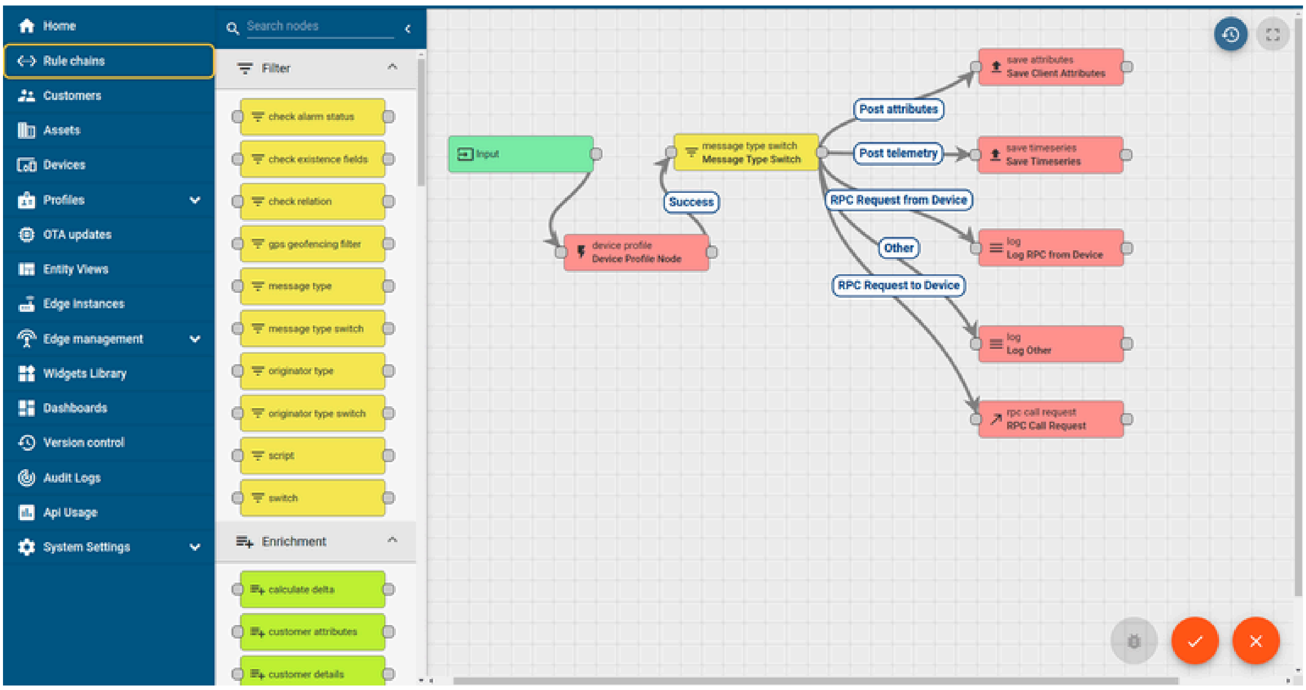
i. **)**

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to• Build Your own dashboard• Analytics and Reporting• Alert and Notification• Integration with third party application(Power BI, SAP, ERP)• Rule Engine





i. Smart Factory Platform (

FACTORY WATCH

i.)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

i.

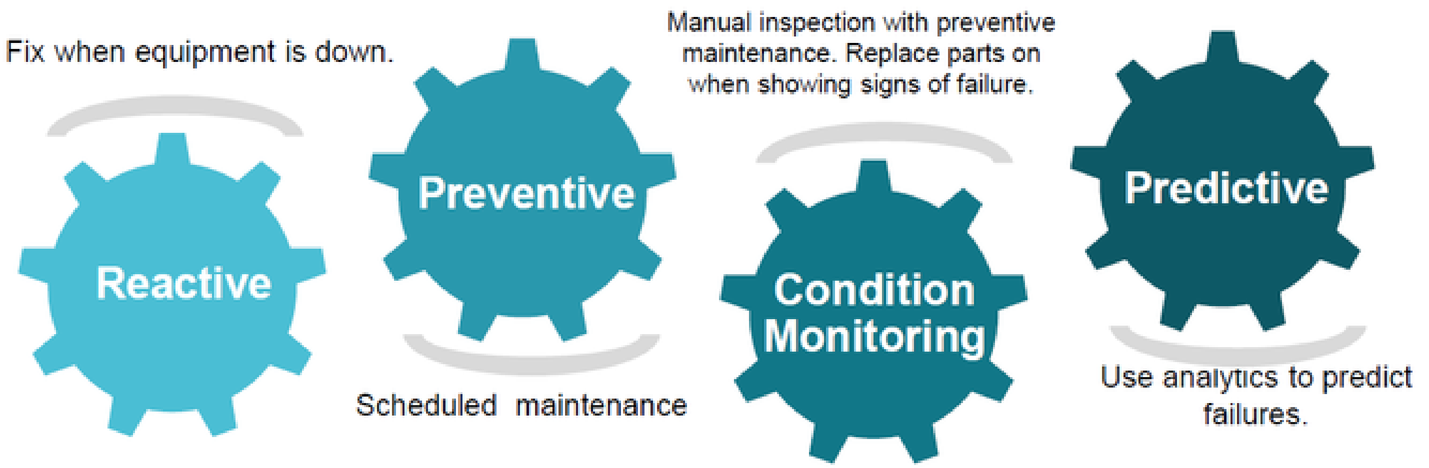
LoRaWAN™

i. based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

i. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

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Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

Reference

[1]

[2]

[3]

Glossary

Terms	Acronym

Problem Statement

Many beginners find it difficult to practice Python programming through real-world applications. A simple interactive quiz application helps learners understand programming concepts while improving logical thinking.

The objective of this project is to develop a Python-based Quiz Game that asks questions, validates answers, calculates the score, and displays the final result in a user-friendly manner.]

Existing and Proposed solution

- Traditional quiz systems require internet access.
- Many applications are complex for beginners.
- Some quiz applications have limited customization options.
- Proposed Solution
- Developed using Python.
- Easy to understand and use.
- Runs offline.
- Calculates score automatically.
- Displays the final result instantly

Code submission (Github link):https://github.com/menakapy/Quiz-Game-

Proposed Design/ Model

High Level Diagram (if applicable)

- Start
- ↓
- Display Welcome Message
- ↓
- Display Questions
- ↓
- Accept User Answer
- ↓
- Check Answer
- ↓
- Update Score
- ↓
- Display Final Score
- ↓
- End

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

Low Level Diagram (if applicable)

- Store questions and answers.
- Display one question at a time.
- Accept user input.
- Compare answer with correct answer.
- Increase score if correct.
- Show final score.

Performance Test

1. **Test plan**

- Verify question display.
- Verify answer validation.
- Verify score calculation.
- Verify final output.

2.Test Procedure

- Run the program.
- Answer all questions.
- Check whether scores are calculated correctly.
- Verify the final result.

3.Perfomance Outcomes

The Quiz Game executed successfully. All questions were displayed correctly, answers were validated accurately, and the final score was generated without errors.

My learnings

During the internship I've learned

The internship improved my confidence in Python programming and software development. Python fundamentals

Variables and Data Types

Conditional Statements

Loops

Functions

Lists

User Input Handling

Debugging

Problem Solving

Logical Thinking

Project Development

Report Writing

Future work scope

The Quiz Game can be enhanced by adding:

- Graphical User Interface (GUI)
- Timer for each question
- Difficulty levels
- Random question generation
- Database connectivity
- User login system
- Leaderboard
- Online multiplayer quiz

Category-wise questions

Score history